

Question Number 1 – Task B.

Question 1 – Return all data for accounts that have no claims in 2024 but at least one claim in 2025.

Answer 1 -

Scan filters

30125525-data-safe-insurance-db

▼ Scan or query items

☒ Scan ☐ Query

Select a table or index: Table - 30125525-data-safe-insurance-db

Select attribute projection: All attributes

▼ Filters - optional

Attribute name	Condition	Type	Value	
Claims Filed 2024	Equal to	Number	0	Remove
Claims Filed 2025	Greater than or equal to	Number	1	Remove

Add filter

Run Reset

Figure 1. Query filters Q1.

Output

Completed - Items returned: 15 - Items scanned: 40 - Efficiency: 37.5% - RCUs consumed: 17.5

Table: 30125525-data-safe-insurance-db - Items returned (15)

Scan started on May 24, 2025, 02:12:56

	Client Username (String)	Client Country (String)	Annual Premium 2024	Annual Premium 2025	Claim Amount 2024	Claim Amount 2025	Claims Filed 2024	Claims Filed 2025
☐	Bernard_T	FR	0	480	0	2300	0	1
☐	Mountain_Tours	US	380	420	0	1228	0	1
☐	Ahmed_Family	UK	720	778	0	4300	0	1
☐	Martinez_R	US	580	625	0	2100	0	1
☐	Lefevre_Family	FR	650	705	5	9410	0	1
☐	Sharma_V	UK	220	231	0	500	0	1
☐	Apex_Group	US	1050	1200	0	6000	0	1
☐	Patel_S	UK	380	405	0	2443	0	1
☐	Petit_Holdings	FR	1150	1250	0	12000	0	1
☐	Lyon_Enterprises	FR	0	1200	0	6000	0	1
☐	Phillips_T	US	0	380	0	1960	0	1
☐	Westbrook_Inc	FR	850	902	0	21033	0	1

Figure 2. Output Q1.

Question 2 – Return Client Username and Country for US accounts with a claim amount of less than £40,000 in 2025.

Answer 2 -

Scan filters

30125525-data-safe-insurance-db

▼ Scan or query items

☒ Scan ☐ Query

Select a table or index: Table - 30125525-data-safe-insurance-db

Select attribute projection: Specific attributes

Specific attributes to project: Client Country, Client Username

▼ Filters - optional

Attribute name	Condition	Type	Value	
Client Country	Equal to	String	US	Remove
Claim Amount 2025	Less than	Number	40000	Remove

Add filter

Run Reset

Figure 3. Query filters Q2.

Output

Completed - Items returned: 8 - Items scanned: 40 - Efficiency: 20% - RCUs consumed: 17.5

Table: 30125525-data-safe-insurance-db - Items returned (8)

Scan started on May 24, 2025, 02:20:05

Client Username (String)	Client Country (String)
Rodriguez_M	US
Mountain_Tours	US
Martinez_R	US
Thompson_K	US
Apex_Group	US
Phillips_T	US
Chang_L	US
Singh_Family	US

Figure 4. Output Q2.

Question 3 – Return the Username for accounts with a Home Policy on the Premium Plan.

Answer 3 -

Scan Filters

The screenshot shows the AWS DynamoDB console interface for the table `30125525-data-safe-insurance-db`. The **Scan or query items** section is active, with the **Scan** option selected. The table `Table - 30125525-data-safe-insurance-db` is selected. The **Select attribute projection** dropdown is set to `Specific attributes`. The **Specific attributes to project** section shows `Client Username` selected. The **Filters - optional** section contains two filters:

Attribute name	Condition	Type	Value	Action
<code>Policy Type</code>	<code>Equal to</code>	<code>String</code>	<code>Home</code>	<code>Remove</code>
<code>Premium Plan</code>	<code>Equal to</code>	<code>String</code>	<code>Y</code>	<code>Remove</code>

Buttons for `Add attribute`, `Add filter`, `Run`, and `Reset` are visible at the bottom of the filter section.

Figure 5. Query filters Q3.

Output

The screenshot shows the AWS DynamoDB console output for a scan operation. The status bar indicates **Completed** with 8 items returned, 40 items scanned, 20% efficiency, and 17.5 RCUs consumed. The table `30125525-data-safe-insurance-db` shows 8 items returned. The first item is `Client Username (String)`.

Client Username (String)
<code>Ahmed_Family</code>
<code>Moreau_Family</code>
<code>Wilson_Family</code>
<code>Westbrook_Inc</code>
<code>Johnson_Family</code>
<code>Chang_L</code>
<code>Weber_Family</code>
<code>Singh_Family</code>

Figure 6. Output Q3.

Question 4 – Return all the data for accounts that have an annual premium of greater than £1000 for both 2024 and 2025.

Answer 4 –

Scan Filter

The screenshot shows the AWS DynamoDB console interface for the table `30125525-data-safe-insurance-db`. The **Scan or query items** section is active, with the **Scan** option selected. The **Select a table or index** dropdown is set to `Table - 30125525-data-safe-insurance-db`, and the **Select attribute projection** dropdown is set to `All attributes`. Under the **Filters - optional** section, two filters are configured:

- Attribute name:** `Annual Premium 2024`, **Condition:** `Greater than`, **Type:** `Number`, **Value:** `1000`. A **Remove** button is present.
- Attribute name:** `Annual Premium 2025`, **Condition:** `Greater than`, **Type:** `Number`, **Value:** `1000`. A **Remove** button is present.

An **Add filter** button is located below the filters. At the bottom, there are **Run** and **Reset** buttons.

Figure 7. Query filters Q4.

Output

The screenshot shows the AWS DynamoDB console interface displaying the results of the scan operation. A green banner at the top indicates the scan is **Completed** with the following statistics: `Items returned: 18`, `Items scanned: 40`, `Efficiency: 45%`, and `RCUs consumed: 17.5`. Below the banner, the table `30125525-data-safe-insurance-db` is shown with the heading **Items returned (18)**. The scan started on May 24, 2025, at 02:30:56. The table lists the following items:

Client Username (String)	Client Country (String)	Annual Premium 2024	Annual Premium 2025	Claim Amount 2024	Claim Amount 2025	Claims Filed 2024	Claims Filed 2025
Mountain_Tours	DE	1800	1950	64000	64100	1	1
Schmidt_GmbH	DE	1400	1512	18000	30600	1	2
Moreau_Family	FR	1300	1411	35000	35153	1	2
Waterston_Group	UK	2400	2550	110000	102335	2	2
Oxford_Properties	UK	3600	3870	120000	236284	1	2
Becker_Systems	DE	2900	3100	140000	148300	2	2
Fischer_Group	DE	2800	2997	75000	153441	1	2
Dubois_Associates	FR	1800	1921	30000	35016	1	2
Apex_Group	US	1050	1200	0	6000	0	1
Martin_Group	FR	1950	2104	8500	8300	1	1
Petit_Holdings	FR	1150	1250	0	12000	0	1

Figure 8. Output Q4.

Question 5 – Identify within the data an account that has an impossible claim amount. Create a filter to detect such accounts for 2024 data.
(Note: One sample within the data has an impossible claim amount).

Answer 5 –

Scan filter

The screenshot shows the AWS DynamoDB console interface for the table '30125525-data-safe-insurance-db'. Under the 'Scan or query items' section, the 'Scan' radio button is selected. The 'Select a table or index' dropdown is set to 'Table - 30125525-data-safe-insurance-db', and the 'Select attribute projection' dropdown is set to 'All attributes'. In the 'Filters - optional' section, two filters are configured:

Attribute name	Condition	Type	Value	Action
Claim Amount 2024	Greater than	Number	0	Remove
Claims Filed 2024	Equal to	Number	0	Remove

Below the filters, there is an 'Add filter' button and 'Run' and 'Reset' buttons at the bottom.

Figure 9. Query filters Q5.

Output

The screenshot shows the 'Table: 30125525-data-safe-insurance-db - Items returned (1)' section. A green status bar at the top indicates 'Completed - Items returned: 1 - Items scanned: 40 - Efficiency: 2.5% - RCUs consumed: 17.5'. The table view shows one item:

	Client Username (String)	Client Country (String)	Annual Premium 2024	Annual Premium 2025	Claim Amount 2024	Claim Amount 2025	Claims Filed 2024	Claims Filed 2025
☐	Lefevre_Family	FR	650	705	5	9410	0	1

Figure 10. Output Q5.